

Resonance Quantum Theory (RQT): A Framework for Resonance-Base Structure and Interaction Conceptual Foundations and the Roton Quantum Model

Olav Le Doigt

June 26, 2026 (v2)

Abstract

The Resonance Quantum Theory (RQT) is a conceptual framework that interprets physical structure, interaction, and persistence as consequences of resonance organization rather than as properties imposed through fundamental particles, fields, or quantization rules. The framework begins from the assumption that localized resonance relationships can form within a continuous resonance-supporting continuum, giving rise to stable structures, interaction channels, and hierarchical organization across multiple scales.

In RQT, observable physical entities are viewed as dynamically maintained resonance structures. Stability arises through resonance closure, while interactions emerge through open and residual resonance channels connecting otherwise localized systems. Information, inertia, and binding are interpreted as consequences of the accommodation and reconfiguration of these resonance relationships. Matter and radiation are therefore treated not as fundamentally distinct categories, but as different manifestations of resonance organization within a common underlying framework.

The theory further proposes that physical structure develops hierarchically. Larger and more persistent systems emerge from networks of mutually stabilizing resonance configurations, while observable forces reflect residual effects of incomplete resonance closure. In this view, quantization, stability, and long-range interaction arise as natural consequences of resonance geometry and organization rather than as independent postulates.

To facilitate visualization, analysis, and numerical investigation, the Roton Quantum Model (RQM) is introduced as a discrete structural representation of the broader RQT framework. RQM projects resonance structures onto simplified resonance centers and channels, allowing complex resonance relationships to be described through a tractable geometric model while preserving the central concepts of closure, hierarchy, and interaction.

The purpose of this paper is to establish the conceptual foundations of RQT, define its principal structural concepts, introduce the relationship between RQT and RQM, and explore how resonance-based organization may provide a unified description of physical structure across scales.

1 Introduction

Nature exhibits persistent structure across an extraordinary range of scales. Atomic systems form stable molecules, molecules organize into complex materials, and larger structures emerge through increasingly intricate patterns of interaction. Despite their diversity, such systems share a common characteristic: they maintain coherence, stability, and recognizable identity while continuously interacting with their surroundings.

Modern physics provides highly successful mathematical descriptions of these phenomena. Quantum theory, quantum field theory, and relativity accurately predict a vast range of experimental observations and have established a remarkably consistent framework for understanding physical behavior. At the same time, many fundamental properties of physical systems are introduced through independent postulates, parameters, symmetries, or interaction rules. Stability,

quantization, binding, charge, and long-range interaction are typically described within specialized theoretical frameworks, each optimized for a particular domain of observation.

This raises a broader conceptual question: might these apparently distinct phenomena arise from a common underlying organizational principle? Rather than beginning with particles, fields, or forces as fundamental entities, one may instead ask whether physical structure itself emerges from the formation and persistence of relationships between interacting systems.

The Resonance Quantum Theory (RQT) explores this possibility. The framework is based on the assumption that resonance relationships can form between localized structures and that these relationships may themselves become stable, persistent, and capable of further interaction. In this view, physical systems are understood not primarily as isolated objects but as dynamically maintained networks of resonance relationships. Stability, interaction, and hierarchy are interpreted as consequences of the organization of these relationships across different scales.

The purpose of RQT is not to replace successful physical descriptions, but to investigate whether a common resonance-based framework can provide a deeper conceptual basis from which a wide range of observed phenomena may emerge. The theory therefore focuses on the formation of resonance structures, the development of stable interaction channels, and the mechanisms through which increasingly complex systems can arise from comparatively simple assumptions.

To facilitate visualization and quantitative exploration of these concepts, a simplified structural representation referred to as the Roton Quantum Model (RQM) is introduced later in this work. RQM provides a discrete geometric description of resonance structures and their interactions while remaining rooted in the broader conceptual framework established by RQT.

The following chapters develop the fundamental concepts of the theory, introduce the notion of resonance channels and closure, and examine how physical structure may emerge from the organization of resonance relationships across multiple levels of complexity.

2 Conceptual Foundations

The conceptual foundation of RQT is intentionally minimal. Rather than beginning with particles, fields, or forces, it starts from the possibility that stable identities may emerge through persistent relationships. Once such an identity has formed, it may itself participate in further relationships, allowing increasingly complex structures to arise through repeated application of the same principle. Figure 1 summarizes this recursive concept, which is developed throughout the remainder of this section.

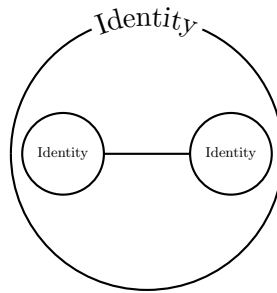


Figure 1: Recursive Formation of Identity

2.1 Persistence and Self-Resonance

Before a structure can interact, it must first persist.

The Resonance Quantum Theory begins with a minimal conceptual assumption: the existence of a continuum capable of supporting oscillatory behavior. The framework does not initially

specify the origin, composition, or geometry of this continuum. It merely assumes that localized oscillatory configurations may arise within it.

An oscillatory configuration does not necessarily constitute a persistent structure. Of particular interest are configurations that repeatedly return to compatible states. Such recurrence allows an oscillation to maintain its identity over extended durations and establishes the possibility of persistence.

Within RQT, persistence is regarded as the first prerequisite for the emergence of structure. A persistent oscillatory configuration forms a self-maintaining cycle whose future evolution remains compatible with its preceding states. This process is referred to as self-resonance. At the conceptual level adopted here, self-resonance may be regarded as the simplest form of persistent resonance behavior.

The precise nature of such self-resonant configurations is not specified at this stage. Their significance lies not in their detailed structure, but in the fact that they provide a mechanism through which stable and persistent patterns may arise. Once persistence is established, increasingly complex forms of coupling, interaction, and organization become possible.

RQT therefore considers persistent self-resonant configurations as the conceptual starting point from which higher-order resonance structures can emerge.

2.2 Stability and Identity

Persistence allows oscillatory behavior to remain. Stability allows it to become structure.

A persistent oscillatory configuration does not necessarily remain unchanged. Local adjustments, interactions, and ongoing accommodation may continuously influence its evolution. Within RQT, stability therefore does not imply rigidity or immutability. Instead, a structure is considered stable when it repeatedly returns to sufficiently compatible states, allowing its characteristic behavior to persist despite ongoing change.

Such stable resonance structures represent the first distinguishable configurations within the framework. Their identity does not arise from a fixed internal composition, but from the persistence of recognizable resonance properties over time. A stable structure may therefore be characterized through its phase relationships, recurrence behavior, and interaction potential rather than through a permanently defined geometry.

The continued existence of a stable resonance structure also preserves information about its formation and evolution. This information is not regarded as a separately stored quantity, but as an intrinsic consequence of the ongoing resonance process itself. The present state of a structure remains connected to preceding states through a continuous resonance flow, allowing distinctions and relationships to persist over time.

Once stable resonance structures emerge, they may participate in increasingly complex forms of coupling and organization. Stability therefore provides the foundation upon which interaction, relationship formation, and higher-order resonance structures can develop.

Within RQT, stable resonance structures constitute the first identifiable building blocks from which larger resonance networks and hierarchical organizations may arise.

2.3 Interaction and Relationship Formation

Interaction creates relationships. Relationships create shared identity.

Stable resonance structures do not exist in complete isolation. Whenever the state of one structure influences the evolution of another, an interaction is established. The precise mechanism through which this influence occurs is not specified at this stage. It is sufficient to recognize that persistent structures may affect one another and thereby participate in a common process of development.

Within RQT, interactions are significant not merely because they modify existing structures, but because they give rise to relationships. A relationship represents an organized connection

between two or more structures whose future evolution can no longer be described entirely independently. Through interaction, participating structures begin to share aspects of their resonance behavior and thereby become part of a larger organizational whole.

The emergence of a relationship introduces an additional level of identity. While the participating structures may remain individually distinguishable, the relationship itself acquires significance as a persistent element of the overall resonance organization. From a higher-level perspective, the relationship may therefore be regarded as a structure in its own right.

Relationships need not possess equal strength or persistence. Some may remain weak and contribute only limited influence on the participating structures. Others may develop increasingly shared resonance organization, causing their behavior to become progressively interdependent. The distinction between weak, strong, and highly persistent relationships is therefore not regarded as fundamental, but as a consequence of the degree to which resonance organization becomes shared.

As relationships persist, they may themselves participate in further interactions and form new relationships with other structures. In this manner, interaction becomes a mechanism through which increasingly complex organizations can emerge. Relationship formation therefore provides the foundation for higher-order resonance structures and the hierarchical organization discussed in the following section.

2.4 Emergence and Hierarchical Organization

Every stable organization becomes a candidate for higher-order organization.

The formation of a relationship does not necessarily result in a new persistent identity. Many relationships may remain transient, while others persist only for limited durations. Within RQT, higher-order structures emerge when relationships acquire sufficient stability to participate in the resonance organization of their surroundings as identifiable entities in their own right.

The persistence of a relationship depends on the degree of compatibility between the participating structures. Compatibility is not regarded as a binary property, but as a condition that may develop through continued interaction. As structures influence one another, adjustments in their relative states may increase or decrease the stability of the relationship. Persistent relationships therefore arise not merely from interaction itself, but from the ability of interacting structures to accommodate one another sufficiently to maintain a shared organization over time.

When a relationship becomes persistent, a new level of identity emerges. This higher-order identity is not simply the sum of its constituents. Instead, it may exhibit resonance capabilities, interaction possibilities, and organizational properties that were not available to the participating structures independently. The emergence of such capabilities constitutes the basis of higher-order structure formation within the framework.

The formation of a new identity may also alter the accessibility of previously available relationships. Certain interactions may become enhanced, while others may become reduced, concealed, or effectively internalized within the newly formed structure. Emergence therefore introduces not only new possibilities for interaction, but also new forms of organization and selective accessibility.

Once established, higher-order identities may themselves participate in further relationship formation. The same principles that govern the emergence of lower-level structures may therefore operate recursively across multiple levels of organization. In this manner, increasingly complex resonance structures can arise without requiring additional foundational assumptions.

RQT consequently interprets hierarchy not as a separate organizing principle, but as a natural consequence of persistent relationship formation, emergent identity, and the continual development of new resonance capabilities across successive levels of organization.

2.5 Accessible and Enclosed Structure

Stability favors enclosure. Enclosure creates selective accessibility.

The emergence of higher-order structures introduces an important distinction between the internal organization of a structure and the properties it exposes to its surroundings. Within RQT, stable structures are not assumed to make all aspects of their internal resonance organization equally accessible. Instead, the formation of persistent identities naturally leads to varying degrees of accessibility between internal and external levels of organization.

This tendency arises from the continuous search for stable resonance configurations. As relationships become organized into higher-order structures, the participating components may reduce their dependence on external interactions by establishing increasingly stable internal relationships. The resulting organization can allow parts of the internal structure to become less directly involved in the structure's interactions with its surroundings.

Enclosure is therefore not regarded as a separate organizing principle, but as a natural consequence of stability. A structure that reduces unnecessary external dependencies may become less susceptible to disturbance and thereby maintain its identity more effectively over time. The degree of enclosure achieved by a structure depends on both its internal organization and the conditions imposed by its environment.

The emergence of enclosure creates selective accessibility. Some properties of a structure remain available for further relationship formation and interaction, while others become partially or fully internalized within the structure itself. As a result, the externally accessible behavior of a structure need not reveal the full complexity of its internal organization.

The accessible properties of a structure define how it participates in higher-order organization. Internal relationships, however, may continue to influence the behavior and stability of the structure even when they are not directly accessible from the outside. In this sense, every emergent structure may possess both an externally visible identity and an internal organization that extends beyond what is immediately available to its surroundings.

RQT therefore regards accessibility and enclosure as complementary aspects of stable organization. The formation of higher-order structures not only creates new identities and interaction possibilities, but also establishes boundaries between what remains externally available and what becomes incorporated into the internal organization of the structure.

2.6 Framework Assumptions and Scope

Complexity is not introduced. Complexity is accumulated.

The preceding sections have established the conceptual foundations upon which the Resonance Quantum Theory is developed. Rather than beginning with particles, forces, fields, or predefined physical laws, the framework starts from a small set of organizational assumptions and explores the consequences that follow from them.

RQT assumes the existence of a continuum capable of supporting oscillatory behavior. Within this continuum, persistent oscillatory configurations may arise and maintain their identity through self-resonance. Persistent structures may influence one another, form relationships, and develop increasingly stable forms of organization. Relationships that become sufficiently persistent may themselves emerge as higher-order identities capable of participating in further relationship formation.

The framework does not assume that all configurations persist. Persistence is itself regarded as a non-trivial property. Only structures that remain sufficiently stable can continue to participate in subsequent organization and relationship formation. Increasing levels of complexity may therefore arise through the continued accumulation of persistent relationships across multiple levels of organization.

From these assumptions emerge the concepts of identity, interaction, hierarchy, accessibility, and enclosure. Higher-order structures are understood not as independent organizing principles, but as consequences of recursive relationship formation and the development of persistent resonance organization.

At this stage, the framework intentionally refrains from specifying the detailed mechanisms

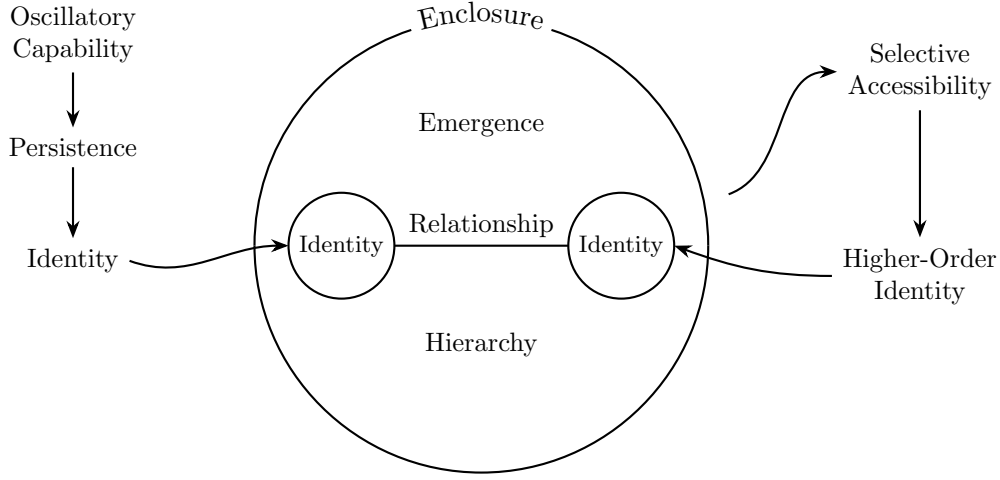


Figure 2: Conceptual foundations of the Resonance Quantum Theory

through which such structures are realized. It does not initially assume particles, forces, charge, mass, spacetime geometry, probabilistic state descriptions, or specific interaction laws. These concepts, if required, are expected to emerge as effective descriptions of deeper resonance processes rather than as fundamental starting points.

The purpose of the framework is therefore not to introduce additional elementary entities, but to investigate how increasingly complex physical organization may arise from a minimal set of resonance-based assumptions. The following sections explore how specific resonance structures, interaction mechanisms, and observable physical phenomena may emerge from these principles.

Future Chapter: Resonance Continuum and Spacetime – Space-time is continuum of perfectly closed resonance channels. Disturbances will open a resonance channel - a Sonon or rather a Sonon-Pair.

Future Chapter: - Prime-Periods: Will they be more stable? - Do we have and indications in nature that prime numbers are favoured or avoided for some parts?

Vibrations – Oscillations – Self-Resonance ...

TODO: Prime periods, factorization hierarchy, resonance richness of composite periods versus prime periods, possible implications for stability and resonance accessibility.

Distance is not necessarily separation in space, but separation and compatibility in identity.

3 The Resonance Quantum Theory

Conceptual foundations of the framework.

3.1 Resonance as a Fundamental Principle

3.2 The Resonance Continuum

3.3 Localized Resonance Structures

3.4 Resonance Accommodation

3.5 Stability and Persistence

4 Resonance Channels and Closure

The formation of observable structure.

- 4.1 Resonance Channels
- 4.2 Open and Closed Channels
- 4.3 Residual Resonance
- 4.4 Resonance Closure
- 4.5 Resonance Enclosure
- 4.6 Resonance Locking
- 5 Information, Stability and Inertia
 - 5.1 Information as Resonance Structure
 - 5.2 Structural Persistence
 - 5.3 Accommodation Dynamics
 - 5.4 Rotonal Inertia
 - 5.5 Hierarchical Stability
- 6 Hierarchical Resonance Structures
 - 6.1 Resonance Nodes
 - 6.2 Resonance Networks
 - 6.3 Resonance Clusters
 - 6.4 Emergent Hierarchies
- 7 The Roton Quantum Model

Discrete representation of resonance structures.

7.1	Purpose of the Model
7.2	Rotons
7.3	Sonons
7.4	Resonons
7.5	Resonance Centers
7.6	Linear Resonance Channels
7.7	Model Limitations
8	Elementary Resonance Structures
8.1	Photons
8.2	Electrons
8.3	Positrons
8.4	Electron-Positron Systems
8.5	Neutrino-like Structures
9	Nuclear Resonance Structures
9.1	Protons
9.2	Neutrons
9.3	Alpha Structures
9.4	Nuclear Resonance Networks
9.5	Binding Energy and Closure
10	Atomic and Molecular Structure
10.1	Atoms
10.2	Molecular Binding
10.3	Resonance Networks
10.4	Closure Hierarchies
11	Interactions and Force Emergence
11.1	Residual Resonance Effects
11.2	Electromagnetic Interaction
11.3	Gravitational Interaction
11.4	Long-Range Resonance Coupling
12	Mathematical Representations
12.1	Continuous Descriptions
12.2	Discrete Descriptions
12.3	Field Approximations
12.4	Numerical Simulation Approaches
13	Predictions and Open Questions

15 Conclusions

Summary and outlook.

Acknowledgements

The author gratefully acknowledges many stimulating interactions with AI-based tools provided by OpenAI, which helped to refine the conceptual structure of the Roton Quantum Model and to clarify its connections to established physical phenomenology. Any remaining inconsistencies or unresolved questions are entirely the responsibility of the author.

A heartfelt thank you is also owed to all the wonderful people who supported this work with their trust and patience, and who accepted the many hours during which the author was not available for everyday activities such as household duties, or shared leisure time.

Author Information

Author: Olav Le Doigt

Profession: Dipl. El. Ing. ETH

ORCID: [0009-0002-3356-9634]

Contact: [olav@ledoigt.ch]

Web: ledoigt.ch

The author is an independent researcher with a long-standing interest in foundational aspects of quantum theory, field dynamics and emergent phenomena in complex systems.

Companion Website and Further Material

A non-technical introduction, extended explanations, graphical illustrations and ongoing discussions of the Roton Quantum Model are available on the project website:

- Roton Quantum Model – intuitive overview and guided tour: <https://ledoigt.ch>
- Technical notes, simulations and updates: <https://ledoigt.ch/ledoigt/notes>
- Discussion and feedback channel: <https://ledoigt.ch/ledoigt/discuss>

These resources are intended as a more accessible companion to the present article, offering step-by-step motivation, visualizations, and evolving refinements of the model.

Data and Code Availability

Simulation scripts, illustrative numerical experiments and example configurations of Rotons used to explore the qualitative behavior reported in this work are (or will be) made available at:

- Code repository (numerical experiments): <https://github.com/ledoigt>
- Supplementary animations and figures: <https://ledoigt.ch/ledoigt/collections/>

Interested readers are invited to reproduce, extend and critically test the presented ideas.

Author’s Note

This work is intended as an open invitation to the theoretical physics and complex-systems communities to examine, challenge and refine the Roton Quantum Model. The author explicitly welcomes critical assessments, alternative formulations and concrete proposals for experimental or numerical tests, which may help to clarify the model’s scope, limits and possible connections to established frameworks.